

NexusFinance — Executive Overview

Digital Lending Platform for Cambodia

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Loan Management • KHQR Payments • Telegram Bot • Android App

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1. Executive Summary

NexusFinance is a full-stack digital lending platform built for microfinance operations in Cambodia. It covers the complete lending lifecycle — from customer loan applications, through officer review and KYC verification, to repayment collection via KHQR payments — with real-time oversight through a secure web dashboard and a Telegram bot.

Why this matters

NexusFinance replaces manual, paper-based lending workflows with a secure, auditable, and bilingual (English / Khmer) digital system.

2. What We Delivered

Capability	Description
Web App	React SPA with three portals — Customer, Loan Officer, Super Admin
Backend API	Node.js/Express REST API with JWT authentication and audit logging
Auth	Email + password with Brevo OTP verification — login blocked until the email is verified
Payments	KHQR (Bakong) integration and ABA PayWay webhooks
Telegram Bot	Payment notifications, reminders, admin commands
Android App	Expo WebView shell — installable APK
Deployment	Vercel (frontend) + Render (backend) + Supabase (database)

3. Demo Accounts

Role	Email	Password
Customer	customer@nexus.com	password123
Loan Officer	officer@nexus.com	password123
Super Admin	admin@nexus.com	password123

Security note

All passwords are stored as bcrypt hashes. OTP codes are hashed and expire after 10 minutes. All admin actions are written to an immutable audit log.

4. Technology Stack

Layer	Technology
Frontend	React 18, TypeScript, Vite, Tailwind CSS
Backend	Node.js, Express, TypeScript (tsx)
Database	Supabase (PostgreSQL)
Auth	JWT + bcrypt + Brevo OTP email
Hosting	Vercel (frontend) + Render (backend)
Payments	KHQR / EMVCo QR, ABA PayWay (sandbox)
Notifications	Telegram Bot API, Brevo transactional email
Mobile	Expo SDK 57, React Native WebView

5. Security & Compliance Highlights

- JWT sessions with server-side role enforcement (customer / loan-officer / super-admin)
- Bcrypt password hashing — no plain-text credentials stored
- Email OTP verification required before first login (new accounts are locked until verified)
- Full audit log with timestamps for every sensitive action
- Rate limiting on auth endpoints to prevent brute force
- Environment variables keep all secrets out of the codebase

6. Roadmap

- ☑ Go live with ABA PayWay production webhooks
- ☑ Add ACLEDA bank integration for payment notifications
- ☑ Connect Vercel auto-deploys (already connected)
- ☑ Add Google OAuth sign-in (deferred)

☑ Launch customer mobile app to app stores